

Week 5 — Effect Modification and Interaction

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: obesity, waist fat and death in MASLD (Kueh et al. 2026)

Two questions hiding in the same 2×2 table

Group	Obese?	Waist fat (WHtR)	People	Deaths
I	yes	lower (< 0.6)	355	4 (1.1%)
II	yes	high (≥ 0.6)	4,265	362 (8.5%)
III	no	lower	732	67 (9.2%)
IV	no	high	559	101 (18.1%)

Question 1 — effect modification: does **waist fat's effect** differ by obesity?

Question 2 — interaction: what do **obesity and waist fat together** do, compared with each alone?

Same table. **Different causal question.** Today we will learn the clean distinction first, then see why the real MASLD example is harder.

Same 5,911 people as Week 4. WHtR = waist-to-height ratio.

The map for today

By the end of today, you should be able to answer **five questions in order**:

1. Am I changing **one exposure** or imagining **two exposures jointly**?
2. What is my modifier M, and could **A change M**?
3. On what **scale** am I comparing effects — additive or multiplicative?
4. What do the **four cells** actually contain?
5. What should I **report**, with uncertainty, without over-interpreting it?

A model term such as $A * M$ comes **after** these questions. It does not define the scientific question for you.

Part 1 · Learn the distinction on a clean example

Part 1 · Learn the distinction on a clean example

Same 2×2 table, two different questions.

Start with smoking, income, and hypertension

Imagine a **fixed follow-up** study:

- ▶ **A = smoking** during the exposure window
- ▶ **M = low income**, measured before that exposure window
- ▶ **Y = incident hypertension** by the end of follow-up

Risk of hypertension	Higher income ($M = 0$)	Low income ($M = 1$)
Not smoking ($A = 0$)	5%	15%
Smoking ($A = 1$)	10%	30%

For this teaching example, assume income is established **before** the smoking exposure window, so smoking cannot move someone between the income strata.

Before any model, read the table in **two ways**: smoking's effect **within income groups**, or the joint pattern across **all four cells**.

Question 1: effect modification

One exposure: smoking. Income defines the groups. Ask whether smoking's effect on hypertension is the same in each income stratum.

	Higher income	Low income
Risk if not smoking	5%	15%
Risk if smoking	10%	30%
Risk difference for smoking	+5 points	+15 points
Risk ratio for smoking	2.0	2.0

- ▶ Additive scale: **yes**, smoking's effect differs (+5 vs +15 points)
- ▶ Multiplicative scale: **no**, the smoking RR is 2 in both income groups

Effect modification is really **effect-measure modification**: the answer can depend on the scale.

In R: read smoking's effect within each income group

```
toy <- data.frame(  
  income = c("Higher", "Low"),  
  risk_nonsmoke = c(0.05, 0.15),  
  risk_smoke    = c(0.10, 0.30)  
)  
  
toy$RD <- toy$risk_smoke - toy$risk_nonsmoke  
toy$RR <- toy$risk_smoke / toy$risk_nonsmoke  
toy  
#>   income risk_nonsmoke risk_smoke   RD RR  
#> 1 Higher          0.05         0.1 0.05 2  
#> 2   Low           0.15         0.3 0.15 2
```

The code does exactly what the table did: compare **smoking vs not smoking within each income group**.

Question 2: interaction

Now make a **stronger move**: treat smoking and low income as **two exposures** and use the higher-income, non-smoking cell as the reference.

- ▶ RR for smoking alone = $10\% / 5\% = 2$
- ▶ RR for low income alone = $15\% / 5\% = 3$
- ▶ RR for both = $30\% / 5\% = 6$

Multiplicative: $6 = 2 \times 3 \rightarrow$ **no multiplicative interaction**

Additive interaction contrast (IC): compare the smoking risk difference across income groups:

$$IC = (R_{11} - R_{01}) - (R_{10} - R_{00}) = R_{11} - R_{10} - R_{01} + R_{00}.$$

Here, $IC = (30 - 15) - (10 - 5) = +10$ **percentage points**. The additive null is **IC = 0**.

The arithmetic is valid, but the **causal interpretation is stronger**. Treating income as a second exposure asks what it means to intervene on income; that may represent several different social interventions. As a modifier, income only had to define a defensible pre-exposure subgroup.

In R: interaction on two scales

```
r00 <- 0.05 # higher income, not smoking
r10 <- 0.10 # higher income, smoking
r01 <- 0.15 # low income, not smoking
r11 <- 0.30 # low income, smoking

rr10 <- r10 / r00
rr01 <- r01 / r00
rr11 <- r11 / r00

mult_ratio <- rr11 / (rr10 * rr01)
IC <- r11 - r10 - r01 + r00
round(c(mult_ratio = mult_ratio, IC = IC), 2)
#> mult_ratio      IC
#>          1.0      0.1
```

$\text{mult_ratio} = 1$ is the multiplicative null. $\text{IC} = 0.10$ means the smoking risk difference is **10 percentage points larger in the low-income group** than in the higher-income group.

Same table, different question

	Effect modification	Interaction
What changes?	Smoking (A) only	Smoking and low income
Role of income (M)	defines pre-exposure groups	second exposure
Main comparison	smoking's effect within each income group	all four joint exposure states
Causal demand	identify smoking → hypertension within income strata	identify joint effects of smoking and income
Scale	additive or multiplicative	additive or multiplicative

Effect modification imagines changing one variable; interaction imagines changing two.

Read more · EpiMethods *Concepts*, “Formal Definitions” · VanderWeele, *Epidemiology* 2009 · Bours, *J Clin Epidemiol* 2021

Confounding is a third idea

	What it means	What you do
Confounding	a common cause biases an estimate	control it
Effect modification	A's effect differs across M	describe stratum-specific effects
Interaction	the joint effect of A and M departs from a chosen scale	describe all four exposure states
Product term $A \times M$	a model parameter	use it as a tool , on that model's scale

A product term is not synonymous with either “effect modification” or “interaction”. The **question and estimand come first**.

Part 2 · A practical workflow for effect modification

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This is the main skill for P3.

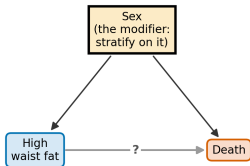
A five-step workflow

1. **Choose A and M.** Give a substantive reason for M.
2. Check that **A cannot cause M**.
3. **Count** people and outcomes in $A \times M$ before modelling.
4. Estimate A's effect within each M on a **named scale**.
5. Report the two effects, their **contrast and uncertainty**, and what you adjusted for.

Do this before asking whether a p-value crossed 0.05.

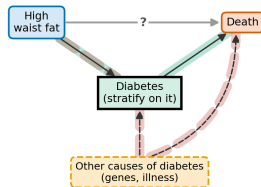
The modifier must survive the “could A change it?” test

A · Fine: the exposure cannot change it



Waist fat cannot change anyone's sex: each stratum holds the same people, whatever their waist size.

B · Not fine: the exposure can change it



Stratifying on diabetes (1) blocks part of waist fat's effect (green) and (2) opens waist fat → diabetes ← other causes → death (red).

Square = stratified on. Dashed outline = not measured. Grey ? = the effect we want. Green = part of the effect; red = an open non-causal path.

“A’s effect among people with $M = m$ ” is cleanest when **A cannot move people into or out of M**. Sex, age, genes, or a condition established before A are natural candidates. A condition measured at the same visit may not be.

If A could cause M, stratifying on M may block part of A’s effect or induce selection/collider bias. That is no longer a simple subgroup effect. In the smoking example, this is why we explicitly defined **income before the smoking exposure window**.

What do we adjust for?

Effect modification of A by M

We need to identify the **effect of A on Y within levels of M**.

- ▶ Control the confounding of **A** $\rightarrow$ **Y** appropriate to that question
- ▶ We do **not** need to identify the causal effect of M itself
- ▶ But conditioning on M still has to be **causally defensible**

Interaction of A and M

To interpret both as exposures, we need to identify the **joint effects of A and M** — a stronger requirement.

The adjustment set follows the **causal question**, not the presence of $A * M$ in software.

Read more · EpiMethods *Concepts*, “Implications for Confounding Control” · Karim, ch. 8

Name the scale before saying “no modification”

Week 4's made-up table, now read for effect modification.

Age group	Risk, exposed	Risk, unexposed	OR	RR	RD
60 or older	0.80	0.50	4.00	1.60	0.30
Under 60	0.50	0.20	4.00	2.50	0.30

- ▶ RR: **modified** by age (1.60 vs 2.50)
- ▶ OR: **not modified** (4.00 vs 4.00)
- ▶ RD: **not modified** (0.30 vs 0.30)

Never write “there is no effect modification” without naming the **scale**.

Two scales are enough for the core lecture

Multiplicative

Compare the **relative effects** of A across M.

For a Cox model:

ratio of HRs = $HR(A \mid M=1) \div HR(A \mid M=0)$

No multiplicative modification $\rightarrow$ **ratio = 1**

For public-health interpretation, the additive scale often answers the concrete question: **how many extra events are associated with the exposure in each subgroup?**

Additive

Compare the **absolute effects** of A across M.

difference of RDs = $RD(A \mid M=1) - RD(A \mid M=0)$

No additive modification $\rightarrow$ **difference = 0**

Part 3 · Now stress-test the idea with MASLD

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The paper is prognostic; our causal re-reading is harder.

The MASLD question we will try to answer

- ▶ **A:** high waist fat, WHtR ≥ 0.6
- ▶ **M:** obesity
- ▶ **Y:** all-cause death
- ▶ **L:** age, sex, race/ethnicity, smoking, sedentary time

Primary Week 5 question: does the effect of high waist fat on death differ between people with and without obesity?

That is an **effect-modification** question: one exposure changes; obesity defines the strata. The paper itself asked a **prognostic phenotype** question; this causal re-reading is ours.

This lecture's analysis: the same 5,911 rows as Week 4, adjusted for L, unweighted (weights are Week 6). Slides that use other rows say so.

Why the MASLD example becomes causally difficult

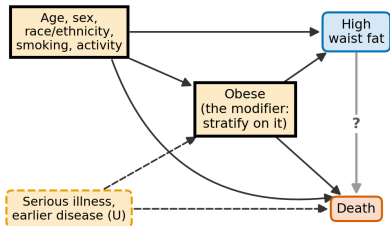
The paper's phenotype analysis is reasonable for **risk stratification**. Extra problems appear when we reuse the same data for a causal effect-modification question:

- ▶ BMI/obesity and WHtR/waist were measured at the **same examination**; the data do not establish which came first
- ▶ both are related measures of adiposity, so treating obesity as clearly upstream of waist fat is a **substantive assumption**
- ▶ the cohort is restricted to people with **MASLD**; MASLD ascertainment depends partly on BMI and waist-related information
- ▶ one important cell is sparse: the paper had **5 deaths** in Group I; our complete-case analysis has **4**

Nothing here makes the paper “wrong.” These complications arise because **we are asking a different, more causal question than the paper was designed to answer.**

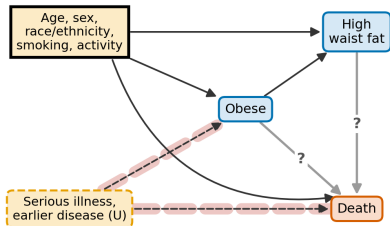
Causal audit: can obesity really be the modifier?

A • Effect modification Does waist fat's effect differ by obesity?



- ✓ {L, obesity} — IF this DAG is right (next slide).
U reaches waist fat only through obesity,
so stratifying on obesity blocks it.

B • Interaction What do obesity and waist fat do together?



- ✗ No valid set, even on this DAG. U confounds
obesity → death (red), and nobody measured it.

Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

Panel A can represent effect modification only under strong assumptions, including that obesity comes **before** waist fat. Panel B treats obesity as a second exposure and exposes an unmeasured confounding problem for obesity → death.

What the effect-modification interpretation assumes

For the obese/not-obese strata to be interpretable as modifiers of waist fat's effect, we need:

- ▶ a time-ordering story in which obesity is established **before** the waist-fat exposure of interest
- ▶ no important unmeasured common cause of obesity and waist fat that becomes problematic when we stratify on obesity
- ▶ L to be adequate for confounding of waist fat → death within each obesity stratum
- ▶ conditioning on MASLD **not to induce important selection bias** in the within-stratum waist-fat comparison

The last assumption is not testable here. FLI, used to identify steatosis, includes **BMI and waist circumference** (plus triglycerides and GGT), so restricting to MASLD can in principle distort the association between adiposity measures and death.

Count first

	Lower waist fat	High waist fat
Obese	Group I: 355 people, 4 deaths	Group II: 4,265 people, 362 deaths
Not obese	Group III: 732 people, 67 deaths	Group IV: 559 people, 101 deaths

- ▶ Group I anchors the obese waist-fat comparison **II vs I** and several phenotype contrasts; it contains only **4 deaths**
- ▶ Among 2,432 obese women, only **57** have lower waist fat: limited support for that exposure level

A model can still produce a coefficient, but the empirical information comes from these cells. **Report the cell counts beside the effect estimates.**

In R: count before you model

```
table(group = d$group, died = d$dead)
#>      died
#> group    0    1
#>  I     351    4
#>  II    3903   362
#>  III    665    67
#>  IV     458   101

with(subset(d, obese == 1 & female == 1),
      table(high_waist = central))
#> high_waist
#>    0    1
#>  57 2375
```

Two lines of code reveal the problem before the Cox model does: **4 deaths in Group I** and very little lower-waist support among obese women.

One product-term model, two stratum-specific HRs

```
fit <- coxph(Surv(time_yr, dead) ~ obese * central +  
            age + female + race + smk_current + sed, data = d)
```

- ▶ Among the **not obese**: HR for high vs lower waist fat = **1.09 (0.79–1.50)**
- ▶ Among the **obese**: HR = **2.99 (1.11–8.09)**
- ▶ Ratio of HRs = **2.74 (0.97–7.71)** (Wald 95% CI)
- ▶ Tests for allowing the HR to differ: **Wald p = 0.06; likelihood-ratio p = 0.03**

$$\exp(\beta_{A \times M}) = \frac{HR_{\text{waist, obese}}}{HR_{\text{waist, not obese}}} = \frac{2.99}{1.09} = 2.74.$$

The product term **re-parameterises the same four cells**. It compares the two HRs; it does not create information where Group I has only four deaths.

In R: recover both HRs and test the product term

```
b <- coef(fit)

round(exp(b[["central"]]), 2)                                # not obese: IV vs III
#> [1] 1.09

round(exp(b[["central"]] + b[["obese:central"]]), 2)        # obese: II vs I
#> [1] 2.99

fit_main <- coxph(Surv(time_yr, dead) ~ obese + central +
                  age + female + race + smk_current + sed, data = d)
round(anova(fit_main, fit)$`Pr(>|Chi|)`[2], 3)              # likelihood-ratio p
#> [1] 0.029
```

The stratum-specific effects are the scientific results; the product-term test is a **contrast between them**.

Why compute standardised six-year risks?

This is **our re-analysis, not a result reported by the paper**. The paper reports Kaplan-Meier curves and Cox hazard ratios. We add standardised risks because effect modification can also be asked on an **absolute, additive scale**.

For a survival outcome we must choose a time horizon. At **6 years**, within each obesity stratum:

1. take the **same people** with their observed age, sex, race/ethnicity, smoking, and sedentary time
2. set everyone's waist-fat exposure to **lower**, predict risk by 6 years, and average
3. set those same people's waist-fat exposure to **high**, predict again, and average
4. subtract the two average risks

A six-year risk is $(1-S(6))$. Standardisation averages those model-predicted risks over the covariate distribution of the people in that obesity stratum.

Additive scale: the standardised six-year results

Obesity stratum	If everyone had lower waist fat	If everyone had high waist fat	Risk difference
Obese	2.3%	6.4%	+4.1 points
Not obese	9.1%	9.8%	+0.7 points

Difference of risk differences = **+3.4 percentage points**

Bootstrap 95% CI: **−0.1 to +6.6 points**

Interpretation: by 6 years, the model predicts about **4.1 extra deaths per 100** with high versus lower waist fat among the obese, versus about **0.7 per 100** among the not obese.

The displayed R code on the next slide calculates the point estimates. The interval comes from a nonparametric bootstrap that refits the model in each resample.

In R: use the same people twice, changing only waist fat

```
# fit is the Cox model from the preceding slides
risk6 <- function(people, waist) {
  nd <- people                # same people, same L values
  nd$central <- waist         # change only high-waist exposure
  s6 <- summary(survfit(fit, newdata = nd), times = 6)$surv
  mean(1 - s6)               # average predicted 6-year risk
}

ob <- subset(d, obese == 1)
notob <- subset(d, obese == 0)

r <- c(
  obese_low = risk6(ob, 0),
  obese_high = risk6(ob, 1),
  notobese_low = risk6(notob, 0),
  notobese_high = risk6(notob, 1)
)
round(100 * r, 1)
#>      obese_low      obese_high notobese_low notobese_high
#>           2.3           6.4           9.1           9.8
```

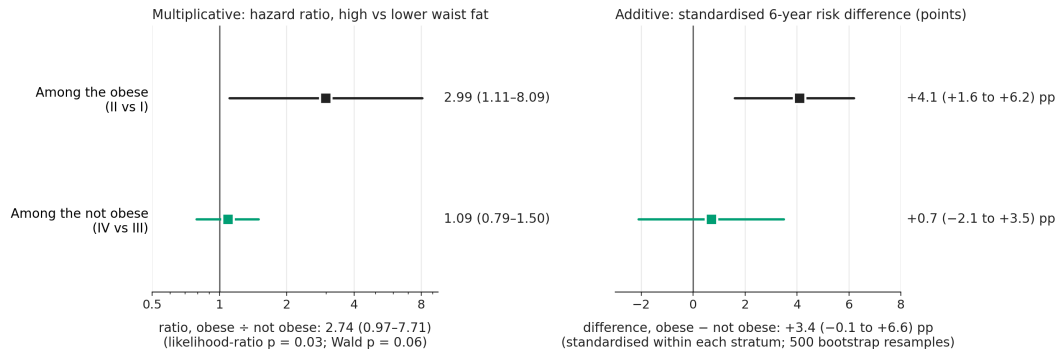
In R: turn the four predictions into risk differences

```
rd <- 100 * c(
  obese      = r[["obese_high"]] - r[["obese_low"]],
  not_obese  = r[["notobese_high"]] - r[["notobese_low"]]
)
round(rd, 1)
#>      obese not_obese
#>      4.1      0.7

round(rd[["obese"]] - rd[["not_obese"]], 1) # difference of risk differences
#> [1] 3.4
```

The code mirrors the estimand: **high minus lower waist fat within each obesity stratum**, then compare those two risk differences.

Put both scales on one page



A reasonable report is: **waist fat may have a larger effect among the obese**, on both the hazard-ratio and six-year risk-difference scales, but the estimate is unstable and depends heavily on a reference cell with only **four deaths**.

The paper's IV vs I comparison asks a different question

Comparison	What changes?	What question does it answer?
II vs I	waist fat only	waist-fat association among the obese
IV vs III	waist fat only	waist-fat association among the not obese
IV vs I	obesity status and waist fat	prognosis of phenotype IV versus phenotype I

Our effect-modification question uses the two **within-row** comparisons: II vs I and IV vs III. The paper's IV vs I contrast is **diagonal** across the 2×2 table.

The paper's IV vs I comparison is not wrong. It is useful for **phenotype risk stratification**. It is simply **not the effect of high waist fat among the not obese**; that contrast is IV vs III.

Part 4 · Interaction, survival tools, and reporting

Part 4 · Interaction, survival tools, and reporting

Now change the question deliberately.

If obesity is a second exposure, the estimand changes

Now deliberately treat both variables as exposures, keeping the coding consistent:

- ▶ **A = high waist fat** (1 = high, 0 = lower)
- ▶ **M = obesity** (1 = obese, 0 = not obese)
- ▶ each person now has four joint potential outcomes, $Y(a, m)$

The interaction contrasts ask whether the waist-fat effect changes when obesity is also changed:

- ▶ **Multiplicative:** ratio of waist-fat HRs = **2.74 (0.97–7.71)**
- ▶ **Additive:** $IC = (R_{11} - R_{01}) - (R_{10} - R_{00})$; using one common standardised population, **IC = +3.9 points**

The arithmetic is estimable, but a **causal** joint-exposure interpretation additionally requires identification of obesity's effect. In our DAG, **U confounds obesity → death and is unmeasured**.

In R: interaction with the same A and M coding

```
# Multiplicative interaction from the Cox product term  
b <- coef(fit)  
round(exp(b[["obese:central"]]), 2)    # HR(waist | obese) / HR(waist | not obese)  
#> [1] 2.74
```

In R: the additive interaction contrast

For the additive joint-exposure contrast, put **all four joint states in the same target population**:

```
# A = high waist, M = obesity
# R00 = Group III; R10 = IV; R01 = I; R11 = II
risk6_all <- function(waist, obese) {  # everyone, set to one joint state
  nd <- d; nd$central <- waist; nd$obese <- obese
  mean(1 - summary(survfit(fit, newdata = nd), times = 6)$surv)
}
R <- c(R00 = risk6_all(0, 0), R10 = risk6_all(1, 0),
      R01 = risk6_all(0, 1), R11 = risk6_all(1, 1))
round(100 * R, 1)
#> R00 R10 R01 R11
#> 6.9 7.5 2.5 7.0

IC <- 100 * (R[["R11"]] - R[["R01"]] - R[["R10"]] + R[["R00"]])
round(IC, 1)  # percentage points
#> [1] 3.9
```

Why **3.9** here but **3.4** for effect modification? Effect modification averaged within each obesity stratum; joint interaction standardises all four exposure states to **one common population**. They are related, but not identical, estimands.

The right survival tool depends on the scale

Goal	Tool	MASLD effect-modification result
Multiplicative	Cox A * M → ratio of HRs	2.74 (0.97–7.71)
Additive	standardised risk at a fixed time → difference of RDs	+3.4 (−0.1 to +6.6) points

- ▶ Do **not** turn a censored survival outcome into “ever died” and fit ordinary logistic regression
- ▶ A fixed-follow-up binary outcome is different; then a binary regression model may be appropriate

For survival outcomes, define the time horizon for an absolute effect: here, **six years**. The standardised risks are **our Week 5 re-analysis**, not quantities reported in the paper.

A dramatic subgroup claim can be built on almost no information

These are **crude diagonal phenotype comparisons** (Group IV vs Group I), fitted separately by sex. They are **not** sex-specific waist-fat effects.

	Men	Women
HR, IV vs I	36.14 (8.86–147.48)	4.11 (0.99–17.12)
Group I deaths / people	2 / 317	2 / 58

The apparent 36-versus-4 gap is dramatic, but each reference survival experience contains only **two deaths**, and both confidence intervals are extremely wide. Do not build a biological story from it.

Crude, unweighted, on the 6,048-row locked domain (dom in the code), so Group I has 375 people here, not the 355 of the 5,911.

In R: reproduce the crude IV-vs-I HRs

```
dom$group <- relevel(factor(dom$group), ref = "I")

fit_men <- coxph(Surv(time_yr, dead) ~ group,
                 data = subset(dom, female == 0))
fit_women <- coxph(Surv(time_yr, dead) ~ group,
                  data = subset(dom, female == 1))

round(summary(fit_men)$conf.int[
  "groupIV", c("exp(coef)", "lower .95", "upper .95")], 2)
#> exp(coef) lower .95 upper .95
#>      36.14      8.86     147.48

round(summary(fit_women)$conf.int[
  "groupIV", c("exp(coef)", "lower .95", "upper .95")], 2)
#> exp(coef) lower .95 upper .95
#>       4.11      0.99      17.12
```

Because Group I is the reference, groupIV is exactly the crude **IV vs I** phenotype HR, calculated separately in men and women.

In R: now count the reference events

```
with(subset(dom, group == "I"), table(female, died = dead))  
#>      died  
#> female    0    1  
#>      0 315    2  
#>      1   56    2
```

Group I contributes only **2 deaths in men and 2 deaths in women**. That is why the sex-specific IV-vs-I HRs are so unstable despite their dramatic point estimates.

Ask the subgroup question cleanly

For the locked waist-fat contrast **IV vs III**, with sex as the modifier (all unweighted):

Model	Ratio of HRs, women $\div$ men	LRT p	Rows
Crude	0.62 (0.26–1.43)	0.278	1,296
Paper's Model 1	1.00 (0.42–2.35)	0.993	1,296
Week 4 pre-exposure set	1.13 (0.48–2.66)	0.78	1,291

Additive six-year RDs are also imprecise: about **+0.8 points in men** and **+0.7 in women** (P3, Model 1).

At this precision there is **no clear evidence of sex modification**, but the interval still allows important differences in either direction.

1,296: P3's paper-replication refits (Groups III + IV complete on the paper's Model 1). 1,291 (168 deaths): Groups III + IV of our 5,911. CIs use Wald; p-values use likelihood-ratio tests.

In R: one test for sex modification on the locked contrast

```
s34 <- subset(d, group %in% c("III", "IV"))
s34$IV <- as.integer(s34$group == "IV")

f0 <- coxph(Surv(time_yr, dead) ~ IV + female +
            age + race + smk_current + sed, data = s34)
f1 <- coxph(Surv(time_yr, dead) ~ IV * female +
            age + race + smk_current + sed, data = s34)

round(summary(f1)$conf.int["IV:female", c(1, 3, 4)], 2) # ratio of HRs
#> exp(coef) lower .95 upper .95
#>      1.13      0.48      2.66

round(anova(f0, f1)$`Pr(>|Chi|)`[2], 2) # likelihood-ratio p
#> [1] 0.78
```

One pre-specified contrast, one product-term test, and an interval wide enough to show the uncertainty.

Honest subgroup reporting

1. **Pre-specify** a plausible modifier and explain why A cannot cause it
2. State the **question**: effect modification or joint interaction
3. **Count** people and outcomes in every $A \times M$ cell
4. Report A's effect within each M on a **named scale**
5. Give the **contrast and interval**, not just a p-value
6. Do not invent a biological explanation for a noisy subgroup difference

Read more · Knol & VanderWeele, *Int J Epidemiol* 2012 · Karim, ch. 8, §8.7–8.8 · Interaction tutorial: effect modification

The algorithm to take into your own paper

1. One exposure or two?

Effect modification = one exposure across groups. Interaction = two exposures jointly.

2. Can A change M?

If yes, simple subgroup effect modification may not be the right estimand.

3. Which scale?

Report multiplicative and additive effects when they answer useful questions.

4. What do the cells contain?

Sparse cells can dominate apparently dramatic interaction estimates.

Key takeaways

- ▶ **Effect modification:** A's effect differs across levels of M; for the usual causal subgroup interpretation, A should not determine M
- ▶ **Interaction:** the joint effect of A and M departs from additivity or multiplicativity
- ▶ The **scale matters:** the same data can modify one measure and not another
- ▶ The **question determines confounding control**; a product term does not
- ▶ With survival outcomes: Cox product term for the multiplicative scale; **standardised risks at a fixed time** for an additive scale
- ▶ In MASLD, the point estimates suggest more waist-fat harm among the obese, but the conclusion is fragile because the obese lower-waist cell anchoring **II vs I has only 4 deaths**

Exit question: name one modifier for your own paper, and give one sentence explaining why your exposure cannot change it.

Appendix · Extensions and diagnostics

Appendix: optional — one model, two parameterisations

```
fit_joint <- coxph(Surv(time_yr, dead) ~ group +  
                  age + female + race + smk_current + sed, data = d)  
fit_prod  <- coxph(Surv(time_yr, dead) ~ obese * central +  
                  age + female + race + smk_current + sed, data = d)  
all.equal(fit_joint$loglik, fit_prod$loglik)  
#> [1] TRUE  
  
round(exp(coef(fit_prod))[c("obese", "central", "obese:central")], 2)  
#>           obese           central obese:central  
#>           0.34             1.09             2.74
```

The four-level factor and product-term parameterisations give the same fitted model.

Appendix: RERI, AP, S, and IC

For a four-cell joint-exposure table with a clearly defined (00) state:

- ▶ **RERI** = $RR_{11} - RR_{10} - RR_{01} + 1$; null = 0
- ▶ **AP** = $RERI / RR_{11}$; null = 0
- ▶ **S** = $(RR_{11} - 1) / [(RR_{10} - 1) + (RR_{01} - 1)]$; null = 1
- ▶ **IC** = $R_{11} - R_{10} - R_{01} + R_{00}$; null = 0

In the core MASLD slides we keep **A = high waist fat** and **M = obesity** throughout. With the four joint states standardised to one common population, the six-year risk-scale **IC = +3.9 points**.

RERI, AP, and S depend on the chosen exposure/reference coding and are not emphasized here because the joint causal effect of obesity is not identified in this example. IC was defined at first use in the smoking-income example.

Appendix: why diabetes is a risky modifier here

```
tab <- table(diabetes = d$t2dm, high_waist = d$central)
round(100 * prop.table(tab, 2))
#>      high_waist
#> diabetes  0   1
#>      0  87 72
#>      1  13 28
```

Diabetes is more common among people with high waist fat (28% vs 13%). That does not prove waist fat causes diabetes, but it makes a same-visit diabetes stratum a poor choice for the simple pre-exposure-modifier interpretation.

Appendix: what splitting by diabetes produces

```
f_no  <- coxph(Surv(time_yr, dead) ~ central + obese + age + female +  
              race + smk_current + sed, data = subset(d, t2dm == 0))  
f_yes <- coxph(Surv(time_yr, dead) ~ central + obese + age + female +  
              race + smk_current + sed, data = subset(d, t2dm == 1))
```

The waist-fat HR is about **1.23** without diabetes and **0.93** with diabetes, but neither number automatically has a clean causal subgroup interpretation if waist fat can affect diabetes.

Where to read more

Topic	EpiMethods and course pages	Key reading
Definitions	<i>Concepts</i> — Formal Definitions · video · slides	VanderWeele 2009; Bours 2021
What to adjust for	<i>Concepts</i> — Implications for Confounding Control	Karim, ch. 8
Scale	<i>Concepts</i> — The Role of the Scale	VanderWeele & Knol 2014
Joint exposure, NHANES	Interaction (joint variable, product term, RERI)	Knol et al. 2011
Worked OR / RR / RD Reporting	Interaction tutorial <i>Concepts</i> — Reporting guideline	Hosmer & Lemeshow 1992 Knol & VanderWeele 2012
Thursday's lab	Exercise · solution	—

References (1/2)

- ▶ Banack HR, Kaufman JS. The obesity paradox: understanding the effect of obesity on mortality among individuals with cardiovascular disease. *Prev Med* 2014;62:96–102.
- ▶ Bours MJL. Tutorial: a nontechnical explanation of the counterfactual definition of effect modification and interaction. *J Clin Epidemiol* 2021;134:113–124.
- ▶ Hosmer DW, Lemeshow S. Confidence interval estimation of interaction. *Epidemiology* 1992;3:452–456.
- ▶ Karim ME. Effect modification in non-randomized studies: methods and applications with propensity scores. Chapter 8 (book chapter).
- ▶ Lajous M, Bijon A, Fagherazzi G, et al. Body mass index, diabetes, and mortality in French women: explaining away a “paradox”. *Epidemiology* 2014;25:10–14.
- ▶ Knol MJ, VanderWeele TJ. Recommendations for presenting analyses of effect modification and interaction. *Int J Epidemiol* 2012;41:514–520.

References (2/2)

- ▶ Knol MJ, VanderWeele TJ, Groenwold RHH, et al. Estimating measures of interaction on an additive scale for preventive exposures. *Eur J Epidemiol* 2011;26:433–438.
- ▶ Kueh MTW, et al. Body weight categories and fat distribution in relation to all-cause mortality among adults with metabolic dysfunction-associated steatotic liver disease. *BMJ Open* 2026;16:e113719.
- ▶ VanderWeele TJ. On the distinction between interaction and effect modification. *Epidemiology* 2009;20:863–871.
- ▶ VanderWeele TJ, Knol MJ. A tutorial on interaction. *Epidemiol Methods* 2014;3:33–72.
- ▶ Westreich D, Greenland S. The Table 2 fallacy: presenting and interpreting confounder and modifier coefficients. *Am J Epidemiol* 2013;177:292–298.

Appendix: the two questions as dagitty code

Paste into the *Model code* box at dagitty.net. With Waist as the only exposure, dagitty returns {**L**, **Obese**}. Mark Obese as an exposure too, and it reports that **no adjustment set exists** — until you remove the latent tag from U.

```
dag {  
  Waist [exposure]  Death [outcome]  U [latent]  
  L -> Obese  L -> Waist  L -> Death  
  U -> Obese  U -> Death  
  Obese -> Waist  Obese -> Death  Waist -> Death  
}
```

L = age, sex, race/ethnicity, smoking, sedentary time. U = serious illness and disease that came before obesity (Week 4's Ill and Before). Then test the assumptions: add G [latent] G -> Obese G -> Waist, or FLI [adjusted] Waist -> FLI Waist -> TG TG -> FLI TG -> Death, and watch every adjustment set disappear.